

Solution Manual

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Alex

Q1 Solution Combo

Suppose Bob starts with 1 credit. Everyday Bob can choose to go to a casino. On Day i (with i starting from 1), he has a $\frac{101-i}{100}$ chance to win his gambling. If he wins, he wins a prize of $\frac{1}{2^i}$ credits. If he loses, then he loses all of the credits accumulated so far, and he has no more money to gamble anymore. Optimally, how many days of gambling should Bob do?

Proposed by Alex Zhan

Answer: $\boxed{4}$

Solution:

We consider the marginal profit from each day. Only when Bob has a positive expected marginal profit will he choose to keep gambling, otherwise the expected amount of money he has will go down and it will not be optimal for him to keep gambling.

On day i :

Expected Revenue: $\frac{101-i}{100} \cdot \frac{1}{2^i}$.

Expected Loss: $\frac{i-1}{100} \cdot (\text{all money accumulated so far}) = \frac{i-1}{100} \cdot (1 + \frac{1}{2} + \dots + \frac{1}{2^{i-1}}) = \frac{i-1}{100} \cdot (2 - \frac{1}{2^{i-1}})$.

Marginal Profit: $\frac{101-i}{100} \cdot \frac{1}{2^i} - \frac{i-1}{100} \cdot (2 - \frac{1}{2^{i-1}}) = \frac{99+i-(i-1)2^{i+1}}{100 \cdot 2^i}$.

The marginal profit is only positive while $99 + i \geq (i-1)2^{i+1} \implies i \leq 4$.

Thus, optimally, Bob should only do 4 days of gambling.

Q2 Solution Number Theory

A number a is n -good if it can be written as $r + bn$ where $0 < r < n$ and $\gcd(r, n) = 1, \gcd(b, n) = 1$. Find the size of the set of distinct remainders mod 3^{200} of perfect squares which are 3^{100} -good.

(*In other words, find $|\{a^2 \bmod 3^{200} | a^2 \text{ is } 3^{100}\text{-good}\}|$.)

Format for answering: If this quantity is expressed in the form of $a \times b^k$, find $a + b + k$.

Proposed by Alex Zhan

Answer: $\boxed{2 \times 3^{198}} \implies 2 + 3 + 198 = \boxed{203}$

Solution:

We first count the number of perfect square residues $\pmod{3^{100}}$ which are coprime to 3^{100} .

First of all, there are $\phi(3^{100}) = 2 \times 3^{99}$ residues which are coprime to 3^{100} .

Note that for any two numbers a and b which are coprime to 3^{100} , we have $a^2 \equiv b^2 \pmod{3^{100}}$ if and only if $3^{100} \mid (a-b)(a+b) \implies 3^{100} \mid a-b$ or $3^{100} \mid a+b$.

Thus, as long as $a \not\equiv b \pmod{3^{100}}$ and $a \not\equiv -b \pmod{3^{100}}$, then a^2 and b^2 have distinct residues $\pmod{3^{100}}$.

Thus, we have exactly $\frac{2 \times 3^{99}}{2} = 3^{99}$ distinct quadratic residues $\pmod{3^{100}}$ which are coprime to 3.

Now, we look for all of the quadratic residues $\pmod{3^{200}}$ which are coprime to 3.

Suppose we have that $a^2 \equiv r \pmod{3^{100}}$ where a is coprime to 3.

Then $(a + k \times 3^{100})^2 \equiv r + 2ak \times 3^{100} \pmod{3^{200}}$.

Note that by varying k over 1 to 3^{100} , we can make $2ak$ cover all possible remainders $\pmod{3^{100}}$. (Remember that $2a$ is coprime to 3^{100}).

What this really means is that for any quadratic residue $r \pmod{3^{100}}$ that is coprime to 3, $r + c \times 3^{100} \pmod{3^{200}}$ is also a quadratic residue.

Which means that for each r , we can find exactly $\phi(3^{100}) = 2 \times 3^{99}$ of the 3^{100} -good residues.

Hence, our final answer is $3^{99} \times 2 \times 3^{99} = 2 \times 3^{198}$.

Also, if you are interested, you can look at a relevant lemma known as Hensel's Lemma which can greatly simplify the process.

Edward

Q1 Solution Algebra

Consider the sequence $\{a_n\}$, where a_n is the n th term in the sequence defined by $a_1 = 1$, $a_2 = 2$ and the recursive relation $a_n = 4a_{n-1} - 4a_{n-2} + 1$ for all $n \geq 3$.

It is possible to define a_n for all $n \leq 0$ requiring the recurrence relation to also be true for $n \leq 2$. In this way, all a_n for $n \leq 0$ can be written as $\frac{a}{2^b}$, where a is odd. Find the sum of the absolute values for the greatest $(-5 > -7)$ 3 n 's where the value of b for n and $n+1$ differs by at least 9.

(For example: $n = -5$ does not satisfy this condition because $a_{-5} = \frac{61}{2^6}$ and $a_{-4} = \frac{59}{2^6}$, and the red and blue numbers (the 6's) do not differ by at least 9)

Answer: $\boxed{1536}$

Solution: First, we need to find a general solution to the inhomogeneous second-order linear recursive relation. To do this, we start by ignoring the inhomoge-

neous part (the red 1) and create a characteristic equation:

$$a_n = 4a_{n-1} - 4a_{n-2} + \textcolor{red}{1} \rightarrow a_n = 4a_{n-1} - 4a_{n-2} \rightarrow t^2 - 4t + 4 = 0$$

This equation has a root of $t = 2$, which gives us 2^n . However, in order to solve it, we need another solution that is linearly independent of this one. (Normally, the other root would give us it, but there is no other root).

In a situation like this, the root we are looking for is usually n times the previous root, so $n2^n$. We can check that this is, in fact, a solution by plugging it into the relation $a_n = 4a_{n-1} - 4a_{n-2}$. (Remember that we are still working with the homogeneous version)

$$\begin{aligned} 4a_{n-1} - 4a_{n-2} &= 4(n-1)2^{n-1} - 4(n-2)2^{n-2} \\ &= (2n-2)2^n - (n-2)2^n \\ &= (2n-2-(n-2))2^n \\ &= n2^n \end{aligned}$$

So, the general solution to the homogeneous equation is a linear combination of the two specific solutions:

$$\lambda_1 2^n + \lambda_2 n 2^n$$

In order to solve the inhomogeneous equation, we have to find a specific solution that satisfies $a_n = 4a_{n-1} - 4a_{n-2} + 1$. It doesn't have to be very complicated, so don't think too hard. The simplest solution in this case is $a_n = 1$. Therefore, we can add this onto our general solution to get our actual solution:

$$\lambda_1 2^n + \lambda_2 n 2^n + 1$$

Now, in order to find the values of λ_1 and λ_2 , we need to list a system of equations using our initial conditions:

$$\begin{cases} \lambda_1 2^1 + \lambda_2 (1)2^1 + 1 = 1 & (n = 1) \\ \lambda_1 2^2 + \lambda_2 (2)2^2 + 1 = 2 & (n = 2) \end{cases}$$

We can get $\lambda_1 = -\frac{1}{4}$ and $\lambda_2 = \frac{1}{4}$, which makes our formula $2^{n-2}(n-1) + 1$ after some simplification.

We can see that for any n , the value of “ b ” is $2 - n - v_2(|n-1|)$, where $v_2(x)$ is the number of 2's in the prime factorization of x . In order for this value to change by at least 9, $v_2(|n-1|)$ must change by at least 8 from n to $n+1$. The value is going to change this much only when we encounter a large power of 2, such as 256. In fact, $n = -256$ is a solution because $2 - (-256) - v_2(|-256-1|)$ is 258 and $2 - (-255) - v_2(|-255-1|)$ is 249.

We can repeat this process to find other ones, and we can see that the values of n are -256 , -512 , and -768 , for a sum of 1536.

Q2 Solution

Find the size (cardinality) of the largest subset $S \subseteq \{1, 2, 3, \dots, 100\}$ such that for any distinct $a, b, c \in S$, $\gcd(a, b, c) = 1$.

Answer: $\boxed{30}$ Solution: We can reinterpret the question as finding a subset of the set of numbers from 1 to 100 where no prime number shows up in the prime factorizations of all the numbers in the subset more than twice. We can include the primes from 50 to 100 because they cannot even show up twice in the prime factorizations. We can also include 1 because it has no primes.

Each time we add a new number, it will have a positive number of distinct prime factors, and because the total number of times primes can appear is 2, this “reduces” the number of uses of primes.

Given this, we should use more prime numbers than composites because a prime number only uses 1 “use”, but a composite number uses 2 or more. Thus, it is optimal that we include all 25 primes from 1 to 100.

To add more numbers, we have to be careful about the value of the numbers under 100, as required in the question. It is very hard to create composite numbers by multiplying larger primes because 11×13 is already greater than 100, so the optimal case is to try to group small primes with larger primes. Using this method we get $86 = 2 \cdot 43$, $93 = 3 \cdot 31$, $95 = 5 \cdot 19$, and $91 = 7 \cdot 13$.

This certainly isn't the only way to add in more composite numbers to the set, but it efficiently uses the prime numbers, so we can check that our solution creates the largest possible size of 30.

Kennan

Q1 Solution

Suppose for a positive number x , we have a quadrilateral with sides $x, 2, 4 - x, 2$ in that order. It contains an inscribed circle of radius x with another inscribed quadrilateral in it that has corresponding parallel edges. Find x .

Proposed by Kennan

First, let the quadrilateral be A,B,C,D and the smaller quadrilateral be E,F,G,H which the parallel lines maps in the same order. Note first that EFGH is cyclic, so we have $\angle EFG + \angle EHG = 180^\circ$. We have parallel sides, so we have $\angle EFG = \angle ABC$, $\angle EHG = \angle ADC$. So we have $\angle ABC + \angle ADC = 180$, and therefore ABCD is cyclic. Notice the two opposite edges are the same length 2, so ABCD is an isosceles trapezoid. Now, we have that the inradius is x . So we have the altitude is $2x$. But the altitude is $\sqrt{4 - (2 - x)^2} = \sqrt{4x - x^2} = 2x$.

We get $4x^2 = 4x - x^2 \implies 5x^2 = 4x$. But x is positive, so $x = \boxed{\frac{4}{5}}$

Q2 Solution

Bob is walking in the cells of a 10×10 grid. Meanwhile, Alice is also walking in the cells of the same 10×10 grid. Alice starts from the bottom-left corner and Bob starts from the top-right corner. Alice walks up or right every minute randomly and Bob also randomly walks down or left every minute. Let the probability that they will collide be $\frac{p}{2^q}$ in simplest form and find q .

Proposed by Kennan

Note that when they collide, they have taken an equal number of steps at the same time point. Since in total they both taken $9 \times 2 = 18$ steps, this means they collide at 9 minutes. Now, note that the number of ways for this to happen is the same as the number of paths in the grid, which is $\binom{18}{9}$, as Alice takes the first half and Bob takes the second half.

There are $(2^9)^2$ total number of ways, since each person has two directions. So after some computation, the answer is $\frac{\binom{18}{9}}{2^{18}} = \frac{12155}{2^{16}}$. So $p = 12155$ and $q = 16$ so the answer is $\boxed{16}$.

This problem could also be a good application of Kummer's or Lucas's Theorem. (These two theorems are used to find $v_p\left(\binom{n}{m}\right)$.) We want to find the number of 2's that will be canceled in $\frac{\binom{18}{9}}{2^{18}}$, which is $18 - \nu_2\left(\binom{18}{9}\right)$. Kummer's Theorem states "The p-adic value of $\binom{n}{m}$ is exactly the number of carries when m is added to $n-m$ in base p ." In this case, $p = 2$, $m = 9$, and $n = 18$, so in binary 9 is 1001. $1001_2 + 1001_2$ carries in the units digit and the 4th digit, so $\nu_2\left(\binom{18}{9}\right) = 2$. This means $q=16$.